
DL for ENG
Designed by Remus Teodorescu 2026

LEARNING OBJECTIVES

What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **read** — a polarisation curve, and name which loss dominates where
- **explain** — the thermoneutral voltage, and why stacks are designed around it
- **state** — why a hot spot in a solid oxide cell is self-reinforcing
- **write** — an objective with a degradation constraint, not just a performance one
- **argue** — why optimisation needs a differentiable model, not just an accurate one
- **recognise** — where the course has been heading since L7

WHAT TODAY ASSUMES

from L10.1	coupled electrochemistry and heat
from L9	convection-diffusion in a channel
from L8	thermal gradients, and why they matter

THE LAST PHYSICS LECTURE

Four of the six rows in the physics table are already familiar. What is new is that the answer is an operating schedule, not a field.

AND A DIFFERENT KIND OF QUESTION

Not what is the best operating point, but what is the best trajectory.

ORIENTATION

Four of Six Rows Are Already Familiar

PHYSICS	IN A SOLID OXIDE CELL	WHERE YOU MET IT
Species transport	convection–diffusion in a channel	L9 — same operator
Charge conservation	elliptic, ionic conduction	L10.1 — same operator
Heat with a source	parabolic, electrochemical source	L8.2, L10.1
Reaction kinetics	Butler–Volmer at the interface	L10.1 — same form
Temperature	600–900 °C	new — and it dominates
Reversibility	same device runs both ways	new

Almost nothing here is new physics. What is new is the operating temperature, which turns thermal gradients from a performance issue into the thing that decides how long the device lives.

What Gets Optimised, and With What

WHAT YOU ALREADY HAVE

- the physics network is L10.1's, with a channel coordinate instead of a particle radius
- what is new is that the network is inside an optimisation, not the end of one
- L6.1's autograd supplies the gradient of the objective, not only of the loss
- and that is the whole argument for a differentiable model

THE NETWORK, CONCRETELY

physics network	channel position and time in, fields out
from L10.1	the same coupled-field structure
from L6.1	autograd, now on the objective
new here	a decision variable, not just a field

THE GRADIENT IS THE POINT

A classical solver evaluates the physics. Autograd differentiates through it, so the operating schedule can be optimised directly.

AND L11.2 DOES THE SAME THING

A robot car instead of a stack, and the same argument for differentiability.

One Device, Two Directions

FUEL ELECTRODE (SOFC MODE)

$$H_2 + O^{2-} \rightarrow H_2O + 2e^-$$

OXYGEN ELECTRODE (SOFC MODE)

$$\frac{1}{2}O_2 + 2e^- \rightarrow O^{2-}$$

- oxygen ions conduct through a dense ceramic electrolyte
- **run it one way and hydrogen and oxygen make electricity** — a fuel cell
- **reverse the current and it splits steam into hydrogen** — an electrolyser
- the fuel electrode is the anode in fuel-cell mode and the cathode in electrolysis

For a PINN this is an unusually good property: a single trained model covers both operating modes, because the sign of the current is an input rather than a different physics.

The Polarisation Curve

NERNST POTENTIAL — THE REVERSIBLE VOLTAGE

$$E = E^0(T) + \frac{RT}{2F}\ln\left(\frac{p_{H_2}\,p_{O_2}^{1/2}}{p_{H_2O}}\right)$$

FUEL CELL

$$V = E - \eta_{act} - \eta_{ohm} - \eta_{conc}$$

ELECTROLYSER

$$V = E + \eta_{act} + \eta_{ohm} + \eta_{conc}$$

- three losses, each dominant in a different current range
- activation kinetics dominate at low current and concentration losses at high
- **note the sign flip** — in fuel-cell mode the overpotentials subtract
- in electrolysis they add
- ohmic loss is current times resistance, so it always rises with current

I–V IN BOTH DIRECTIONS

Current density on the horizontal axis. Positive is electrolysis, negative is fuel-cell mode; the curve passes smoothly through the open-circuit voltage at zero.

The Thermoneutral Voltage

WHERE ELECTRICAL INPUT EXACTLY MATCHES THE REACTION ENTHALPY

$$V_{tn} = \frac{\Delta H}{2F} \approx 1.29\,V \quad (steam,\ 800\,^{\circ}C)$$

NET HEAT GENERATION

$$q = i(V - V_{tn})$$

- below the thermoneutral voltage the cell absorbs heat and must be warmed
- above it, the cell generates heat and must be cooled
- most electrolyser stacks are designed around that point
- the heat source has an unusually elegant form
- one expression replaces the four-term source of L10.1

THREE OPERATING REGIMES

$V < V_{tn}$ **endothermal**

heat must be supplied; efficiency above 100% electrical; lowest degradation

$V = V_{tn}$ **thermoneutral**

net heat flux zero; the design point for most stacks

$V > V_{tn}$ **exothermal**

cell must be cooled; higher production, faster ageing

SOEC operating practice; ISPT (2023)

Gas Transport Along the Channel

SPECIES TRANSPORT ALONG THE FLOW DIRECTION

$$\frac{\partial c_k}{\partial t} + u\frac{\partial c_k}{\partial x} = D_k\frac{\partial^2 c_k}{\partial x^2} + S_k(i)$$

REACTION SOURCE

$$\dot{n}_{H_2} = \frac{iA}{2F}$$

- convection dominates in the gas channel; diffusion matters in the porous layers
- composition changes along the channel, so the Nernst potential does too
- the applied voltage is essentially uniform; it is the current density that varies
- which is why a zero-dimensional model is a starting point, not an answer
- reactant utilisation is the design variable

FORMULATION

The Coupled System

CHARGE CONSERVATION — ELLIPTIC

$$\nabla \cdot (\sigma_{ion} \nabla \phi) = 0, \qquad i = -\sigma_{ion} \nabla \phi$$

RESISTANCE FALLS SHARPLY WITH TEMPERATURE

$$ASR(T) = ASR_0 \exp\left[\frac{E_a}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)\right]$$

ENERGY CONSERVATION WITH THE ELECTROCHEMICAL SOURCE

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\lambda \nabla T) + i(V - V_{tn})$$

- two heat sources: ohmic, from resistance to current, and electrochemical, from the reactions
- and the coupling is stronger here than in a battery
- the resistance is Arrhenius in temperature, so a hot spot draws more current
- which makes it hotter
- in a battery a gradient degrades performance; here it cracks ceramic

Start With the Simplest Model

BUTTON CELL

0-D

Uniform everything.
One algebraic voltage balance plus a lumped temperature.
Fits published polarisation data.
Where the exercise starts.

CHANNEL

1-D

Composition varies along the flow; current density with it.
Adds the L9 transport operator.
Captures utilisation effects.
The realistic teaching model.

REPEAT UNIT

2-D / stack

Cross-flow or counter-flow, with in-plane thermal gradients.
This is where thermal stress and lifetime actually live.
A stretch target.

- the button-cell model is not a toy
- a zero-dimensional model is a modelling choice, not a size of cell
- so it is what you can actually compare against
- fit it to a published curve first

VERIFICATION

What to Check, and Against What

POLARISATION CURVE

The primary check. Published i–V curves exist for button cells at stated temperature and gas composition — reproduce one before believing anything else.

OPEN-CIRCUIT VOLTAGE

At zero current the model must return the Nernst potential for the stated composition. A pure thermodynamic check, with no kinetics involved.

THERMONEUTRAL POINT

The voltage at which net heat generation crosses zero should come out near 1.29 V for steam at high temperature. If not, the enthalpy or the sign convention is wrong.

TEMPERATURE SENSITIVITY

Repeat the polarisation curve at two temperatures and check that the ohmic slope changes by the Arrhenius factor. This tests the coupling, not just the steady state.

Unlike the battery half of L10, there is no PyBaMM here — no single community reference implementation. That makes these four checks the whole of your verification, and it makes stating your data source non-negotiable.

DEGRADATION

What Actually Kills a Solid Oxide Cell

A USABLE EMPIRICAL DEGRADATION LAW

$$\frac{d\,ASR}{dt} = k_0 \exp\left(-\frac{E_d}{RT}\right)|i|^m$$

- degradation shows up as a rising area-specific resistance
- **the mechanisms are microstructural** — nickel coarsening, delamination, chromium poisoning
- the published numbers span an enormous range
- the law below is a teaching assumption; real degradation is rarely linear in time
- many factors drive it, so treat any single expression as indicative only

DEGRADATION GROWS WITH T

Illustrative Arrhenius shape against temperature in °C. Higher temperature generally accelerates degradation, though many other factors contribute.

The Trade-Off That Makes This an Optimisation Problem

BOTH MOVE THE SAME WAY WITH TEMPERATURE — AND THAT IS THE WHOLE PROBLEM

$$\dot{n}_{H_2} \uparrow \ \text{with } T, \qquad \frac{d\,ASR}{dt} \uparrow \ \text{with } T$$

- raise the temperature and resistance falls, so the cell consumes less energy now
- but degradation generally accelerates, so it produces less later
- reducing operating temperature is a recognised strategy for slowing that
- the trade-off is better stated in current density: higher density, more output, more degradation
- so the question is not the best operating point but the best trajectory

PULLING IN OPPOSITE DIRECTIONS

operating temperature → (schematic, not measured)

Writing Down What You Actually Want

VALUE PRODUCED OVER THE LIFE OF THE DEVICE

$$\max_{i(t),\,T(t)} \int_0^{t_{life}} \left[p_{H_2} \dot{n}_{H_2} - c_e\, iAV \right] dt$$

SUBJECT TO END-OF-LIFE AND THERMAL-STRESS LIMITS

$$\text{subject to} \quad ASR(t_{life}) \le ASR_{max}, \qquad |\nabla T| \le |\nabla T|_{max}$$

- **two terms in the objective** — hydrogen is worth something, electricity costs something
- two constraints that matter
- the resistance must not exceed its end-of-life threshold before the design lifetime
- and the thermal gradient must stay below what the ceramic tolerates
- the second is what makes a distributed model necessary

Why a Differentiable Model Changes the Problem

RECEDING-HORIZON CONTROL WITH A LEARNED MODEL

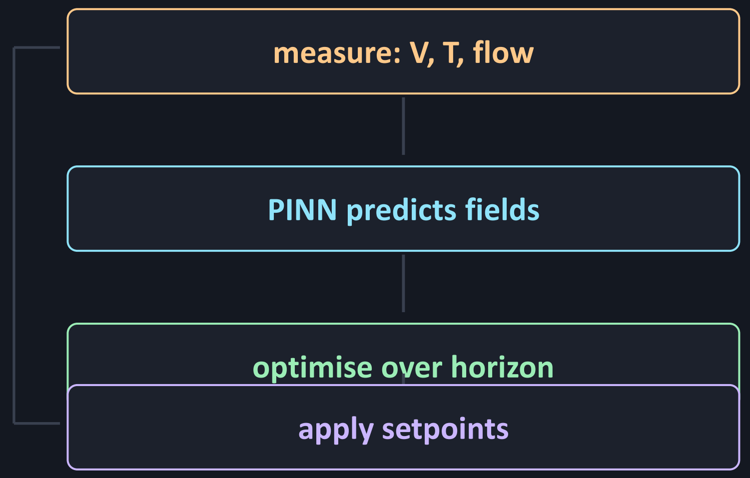
$$\min_{u_{0..N}} \sum_{k=0}^{N} \ell(x_k, u_k) \quad \textit{with } x_{k+1} = \mathcal{N}_{\widehat{w}}(x_k, u_k)$$

AND THE GRADIENT IS FREE

$\frac{\partial J}{\partial u}$ *from the same autograd*

- a classical solver evaluates the physics; to optimise through it you finite-difference or adjoint
- a PINN is differentiable by construction
- so the gradient of the objective with respect to the operating schedule is available directly
- that is the difference between optimising offline and optimising in the loop
- and it makes a constraint on the thermal gradient tractable

THE CONTROL LOOP



What Each Predicted Field Buys You

PREDICTED FIELD	FROM	WHAT IT DECIDES
Temperature field	L8, L10.1	the degradation rate, and the thermal-stress constraint
Thermal gradient	L8.1	the crack criterion — a lumped model cannot express it
Species concentration	L9, L10.2	local Nernst potential and the utilisation limit
Flow field	L9.1	how heat and reactant are actually distributed
Resistance ASR(t)	L10.2	state of health, and how far from end of life

Read the middle column. Every lecture in this course produces one row of this table, and the optimisation is what consumes them all at once.

LIMITS

Where to Be Careful

Degradation laws are empirical	The Arrhenius-times-power form is a fit, not a mechanism, and published rates vary by orders of magnitude.	Never present a predicted lifetime as a number. Present a trend, and state the fitted law.
Extrapolating in time	A model fitted to 1000 hours of data used to predict 40 000 hours of operation.	The dominant mechanism can change with age — early degradation often differs from steady behaviour.
Optimising a wrong model	A gradient-based optimiser will exploit any error in the surrogate, confidently.	Constrain the optimiser to the region where the model was validated, and say where that is.
No community reference	Unlike PyBaMM for batteries, there is no single trusted open implementation to check against.	Cite the exact source of every parameter and every data set you fit to.
Thermal stress is structural	Cracking depends on mechanical properties this model does not contain.	A gradient limit is a proxy. Treat it as a screening criterion, not a failure prediction.

The third is the one that bites in practice: an optimiser is an adversary that finds your model's weakest point and operates there.

OVER TO YOU

Questions

TEN TO TEST WHETHER TODAY LANDED

- which loss dominates at low current, and which at high?
- why do the overpotentials add in electrolysis and subtract in fuel-cell mode?
- what is special about the thermoneutral voltage?
- why does composition change the Nernst potential along a channel?
- why is a hot spot self-reinforcing here and not in a battery?
- what does degradation depend on, and with what shape?

AND FOUR MORE

seven	why is a button-cell model a reasonable starting point?
eight	what makes the thermal-gradient constraint need a distributed model?
nine	why does optimisation want a differentiable model?
ten	what has this course been building towards?

THE ONE WITH NO SETTLED ANSWER

Degradation rates span an order of magnitude. What do you optimise against?

BEFORE L11.1

Ex_10.2 notebooks 01 and 02. Bring your operating trajectory.

EXERCISE

Ex_10.2 — From Button Cell to Lifetime Optimisation

TASKS

1. Fit a 0-D button-cell model to a published polarisation curve and verify the open-circuit voltage.
2. Locate the thermoneutral point and confirm that net heat generation crosses zero there.
3. Extend to a 1-D channel and show how utilisation shifts the local Nernst potential.
4. Add the degradation law and simulate resistance growth at three operating temperatures.
5. Optimise the operating trajectory against a time-varying electricity price, subject to a lifetime constraint.

QUESTIONS — AND WHERE THIS DECK ANSWERS THEM

Why is the thermoneutral point special?	sl. 5
Why does a lumped model fail for lifetime?	sl. 12
What makes temperature a trade-off rather than a setting?	sl. 11
Why does differentiability change what is possible?	sl. 13
What would you refuse to claim from your result?	sl. 15

The last question carries as many marks as the first four. Knowing what your model cannot support is the skill this course is really teaching.

SUMMARY

Key Takeaways

1 **Same physics, hotter**	Species transport, charge conservation and heat — the operators of L9 and L10.1. What changes is 800 °C, which makes thermal gradients the thing that ends the device's life.
2 **One device, two directions**	Fuel cell and electrolyser are the same hardware with the current reversed, so one trained model serves both. The sign of the current is an input, not a different physics.
3 **Thermoneutral is the design point**	At V_{tn} the net heat flux is zero and stack thermal management becomes trivial. Everything above it is production bought with heat and with life.
4 **Lifetime is the real objective**	Temperature raises both production and degradation. There is no best setting, only a best trajectory — and it depends on prices that change hourly.
5 **Differentiability is the point**	A classical solver evaluates the physics; a PINN differentiates it. That is what moves optimisation from offline scenarios to online control.

Next — L11: robotics, where the physics-informed idea survives without a governing PDE at all.

REFERENCES

Sources and Where to Look Next

Thermoneutral operation	SOEC stacks are typically operated at the thermoneutral voltage for steam splitting, about 1.29 V, against roughly 1.48 V for low-temperature water electrolysis — a stack-level efficiency gain of order 15%.
Temperature and degradation	Long-term SOEC tests consistently report faster degradation at higher operating temperature. Lowering temperature is a recognised mitigation, at the cost of production rate — the trade-off this lecture is built on.
Long-duration benchmarks	Multi-thousand-hour cell and stack tests exist in the open literature, with degradation traced by impedance spectroscopy. These are the data sets to fit against.
Gu & Wang (2000)	The thermal-electrochemical coupling method of L10.1 transfers directly: the heat source, the Arrhenius property dependence, and the argument against decoupled models.
Find the current numbers yourself	Degradation rates and cell performance move quickly. Locating and citing the state of the art is part of the exercise — as it was for turbulence in L9.2.

No PyBaMM equivalent exists for solid oxide cells, so every parameter and data set must be cited explicitly.